Evaluating the Groundedness and Robustness of Natural Language Autoencoders

1 Introduction

Understanding what information a language model encodes in its internal activations is a central goal of interpretability research (Belinkov, 2022). Natural Language Autoencoders (NLAs) approach this problem by attempting to translate high-dimensional activations directly into human-readable text explanations (Fraser-Taliente et al., 2026). An NLA consists of two learned components: an encoder, called the *activation verbalizer* (AV), which maps an activation vector to a natural-language explanation; and a decoder, called the *activation reconstructor* (AR), which maps the AV's explanation back to activation space.

Specifically, the AV is an LLM that takes an internal activation vector $h_l \in \mathbb{R}^{d_{\text{model}}}$ from a specific layer of the target model and autoregressively samples a natural-language description z. The AR is a truncated copy of the target LLM with a learned affine projection applied to the final token's layer activation, attempting to reconstruct $\hat{h}_l = \text{AR}_\theta(z) \in \mathbb{R}^{d_{\text{model}}}$ from the explanation z.

The AV and AR are initialized with a "warm start": for roughly 500,000 text snippets, a teacher LLM guesses what information a model would use to predict the next tokens, and the AV is trained to produce that guess from the activation (Fraser-Taliente et al., 2026). Afterward, the AV and AR are trained to minimize the mean-squared reconstruction loss

$$\mathcal{L}(\phi, \theta) = \mathbb{E}_{h_l \sim \mathcal{H}} \mathbb{E}_{z \sim \text{AV}_\phi(\cdot|h_l)} \left\| h_l - \text{AR}_\theta(z) \right\|_2^2 . \tag{1}$$

Here, $\mathcal{H}$ denotes the distribution of target-model activations. This objective rewards explanations that preserve information about the original activation. Reconstruction quality is evaluated using the fraction of variance explained (FVE) (Fraser-Taliente et al., 2026):

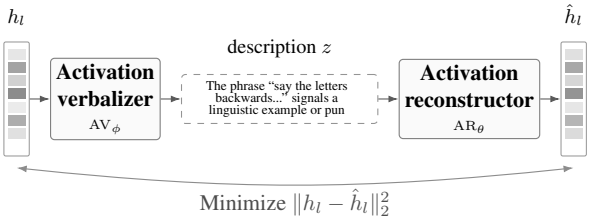


Figure 1: Overview of a Natural Language Autoencoder. The activation verbalizer maps an internal activation h_l to a natural-language explanation z, and the activation reconstructor maps z to a reconstructed activation $\hat{h}_l$. Training minimizes reconstruction error between the original and reconstructed activations.

$$\text{FVE} = 1 - \frac{\mathcal{L}(\phi, \theta)}{\mathbb{E}_{h_l \sim \mathcal{H}} \left[\left\| h_l - \bar{h}_l \right\|_2^2 \right]}, \qquad n\bar{h}_l = \mathbb{E}_{h_l \sim \mathcal{H}}[h_l]. \tag{2}$$

The AV is regularized with a KL-divergence penalty toward the warm-started model. This helps discourage the AV from drifting toward an uninterpretable shorthand code (Fraser-Taliente et al., 2026).

Because this training objective is unsupervised, the AV can, in principle, capture arbitrary latent information encoded within an activation vector, including features that researchers did not anticipate. This makes NLAs appealing as tools for open-ended safety analysis. Recent pre-deployment audits of frontier models, for example, have used NLA explanations to identify unverbalized evaluation awareness and diagnose anomalous behavior caused by malformed training data (Fraser-Taliente et al., 2026).

However, the mathematical and empirical properties of NLAs are not yet well understood. Two questions are particularly salient.

First, are AV explanations **grounded** in the information actually encoded in h_l? Because the AV is itself an LLM with extensive parametric knowledge and reasoning capabilities, it may confabulate highly plausible and coherent claims that are not

supported by the activation itself (Fraser-Taliente et al., 2026; Jacovi and Goldberg, 2020). Low reconstruction error is compatible with this kind of confabulation: the AR may recover h_l from generic or statistically likely features of the AV explanation, rather than from information specific to the original activation. Moreover, the warm-start procedure may actively promote confabulation: it trains the AV to produce plausible descriptions of what h_l might contain, rather than descriptions of what it does contain. Verifying groundedness is challenging, because we lack ground truth for what information an activation contains at the level of specificity expressed by an NLA explanation.

Second, is the AV–AR pipeline **robust**? If explanations vary substantially under small perturbations to the activation, the interpretation prompt, or the sampled explanation z, then any individual explanation may reflect incidental features or noise in the experimental setup rather than stable properties of the target model (Alvarez-Melis and Jaakkola, 2018).

Establishing these properties is essential if NLA explanations are to support safety-relevant conclusions: a plausible, yet unfaithful or unstable explanation may mislead researchers into over-trusting the target model, or into forming spurious hypotheses about its behavior.

2 Groundedness

When the AV makes a claim about the contents of an activation h, how can we determine whether that claim is grounded in information actually encoded in h? We distinguish three possible sources of an AV claim:

Grounded. The activation h contains information that supports the specific claim over plausible alternatives with the same broader theme.

Entangled. h contains information specific to the claim, but that information cannot be uniquely attributed to the claim's recovered direction because other real claim directions have comparable effects.

Seeded. h supports the broader theme of the claim, but it does not contain enough information to distinguish the claim from other theme-consistent alternatives. For example, h may encode "US legislation," while the AV invents a plausible but unsupported detail such as "Senate filibuster."

Confabulated. h supports neither the specific claim nor its broader theme; the AV hallucinates the claim.

Unresolved. The experiment does not support a reliable classification.

2.1 Overview of the Cycle-Consistency Test

We evaluate whether individual claims are grounded using a *cycle-consistency causal grounding test*. For each claim C_i in an AV explanation, we first estimate an activation-space direction associated with C_i. Because the AR maps natural-language explanations back into activation space, we obtain this direction by comparing the AR's reconstruction of the complete explanation with its reconstruction of the same explanation after C_i has been deleted.

We then remove the component of the original activation h along this direction and pass the resulting activation back to the AV. Finally, we measure how much the intervention reduces the AV's score for C_i. A negligible reduction suggests that the AV did not rely on the recovered direction when generating C_i, and therefore that C_i was confabulated.

A large reduction, however, does not distinguish a grounded claim from a seeded one. Because the recovered direction is derived from the text of C_i, it may encode both the claim's specific content and the explanation's broader theme. If h contains only the broader theme, removing this direction can still weaken the thematic signal and thereby reduce the probability of C_i in the AV's explanation. Moreover, ablating C_i's direction may push the activation away from a representation that supports C_i's tokens, regardless of whether the specific content of C_i was originally encoded in h.

We control for these effects using matched *decoy claims* (cf. Hewitt and Liang, 2019). For each real claim C_i, we construct a set of theme-matched claims $C^*_{i,1}, \ldots, C^*_{i,k}$ that the AV did not produce, insert each decoy into the same position in the explanation, and require that its specific content not be supported by the source text. We then apply the same self-ablation procedure to each decoy. These decoy effects provide a baseline for how much a claim's score can fall simply because the intervention removes shared thematic information or otherwise perturbs the activation in a way that disfavors the claim. We classify C_i as grounded only

when its self-ablation effect substantially exceeds this matched-decoy baseline.

2.2 Constructing Claim Directions

We first partition the explanation z into atomic claims $C_1, \ldots, C_n$, where each claim expresses a single assertion about the source. For each claim C_i, let z_{-i} denote the explanation obtained by deleting exactly the token span corresponding to C_i. We then construct a corresponding direction in activation space using the AR:

1. We pass the complete explanation z and the claim-deleted explanation z_{-i} through the AR.

2. We L_2-normalize both reconstructed activation vectors:

$$v_z = \frac{\mathrm{AR}_\theta(z)}{\|\mathrm{AR}_\theta(z)\|_2}, \qquad v_{-i} = \frac{\mathrm{AR}_\theta(z_{-i})}{\|\mathrm{AR}_\theta(z_{-i})\|_2}. \tag{3}$$

3. We subtract the reconstruction without C_i from the reconstruction with C_i:

$$\Delta_i = v_z - v_{-i}. \tag{4}$$

   This difference isolates the change in the normalized AR reconstruction caused by including the text of C_i.

4. Finally, we normalize this difference to obtain the unit claim direction:

$$\widehat{\Delta}_i = \frac{\Delta_i}{\|\Delta_i\|_2}. \tag{5}$$

Thus, $\widehat{\Delta}_i$ points from the reconstruction of z_{-i} toward the reconstruction of the complete explanation z. It represents the activation-space direction that the AR attributes to the inclusion of claim C_i. Under approximate geometric alignment between the AV and AR, $\widehat{\Delta}_i$ is a candidate direction that the AV may have read from the original activation when generating C_i.

2.3 Causal Ablation and Re-Reading

To test whether information along the recovered direction $\widehat{\Delta}_i$ was causally used by the AV to generate C_i, we project this direction out of the original activation (cf. Elazar et al., 2021):

$$h_i^{\mathrm{abl}} = h - \left(h^\top \widehat{\Delta}_i\right) \widehat{\Delta}_i. \tag{6}$$

This intervention removes only the component of h aligned with $\widehat{\Delta}_i$.

Next, we test whether this intervention reduces the AV's support for C_i. A natural approach is to sample a new explanation from the ablated activation h_i^{abl} and check whether C_i still appears. However, this procedure is highly unreliable and noisy: autoregressive generation is stochastic, and an early token change can propagate through the rest of the explanation for reasons unrelated to C_i. It also reduces the effect to a binary outcome—whether C_i appears—rather than measuring the change in the AV's probability of producing the claim.

Instead, we re-score the original explanation under the AV. Let $z = (z_1, \ldots, z_T)$ denote the explanation originally sampled from h. We run the AV once with h and once with h_i^{abl}, while in both cases teacher-forcing the same original explanation. At each position t, we compare the probability assigned to the original token z_t while holding its left context $z_{<t}$ fixed. The two forward passes differ only in the injected activation.

For a claim C_j, let I_j denote the set of token positions belonging to that claim. We define its length-normalized score under activation x as

$$S_z(C_j \mid x) = \frac{1}{|I_j|} \sum_{t \in I_j} \log p_{\mathrm{AV}}(z_t \mid z_{<t}, x). \tag{7}$$

We then measure how much ablating the direction associated with C_i changes the score of C_j:

$$\Delta \log p_{i \to j} = S_z(C_j \mid h) - S_z(C_j \mid h_i^{\mathrm{abl}}). \tag{8}$$

A positive value means that removing $\widehat{\Delta}_i$ makes C_j less likely under the AV, whereas a value near zero means that the AV's support for C_j is largely unchanged by the intervention.

The special case $j = i$ is the claim's *self-ablation effect*; how much removing the direction recovered from C_i reduces the AV's score for C_i itself:

$$R_i \equiv \Delta \log p_{i \to i}. \tag{9}$$

A small self-ablation effect suggests that C_i is confabulated, since the AV's support for the claim is largely unchanged when its activation-space direction is removed.

2.4 Matched Decoy Controls

A positive self-ablation effect does not necessarily establish that the specific content of C_i is grounded. There are two important ways this can happen even when h does not encode the claim itself.

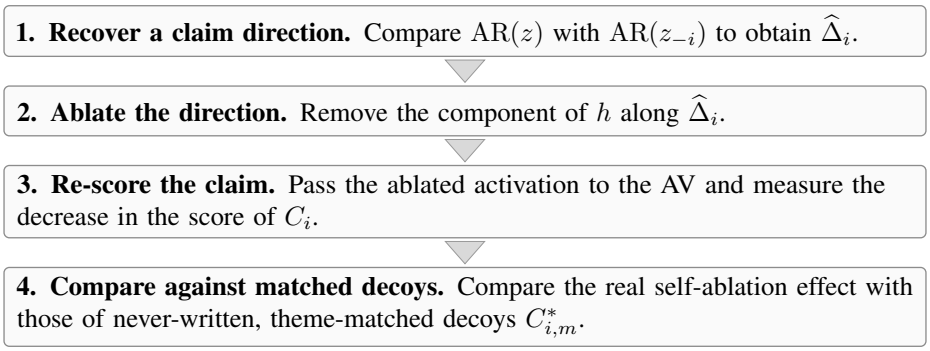


Figure 2: High-level overview of the cycle-consistency causal grounding test.

Theme piggybacking. The direction $\widehat{\Delta}_i$ is recovered from the *text* of C_i, so it may contain both the claim's specific content and the broader theme that makes the claim plausible. Suppose, for example, that h encodes only "Korean history," while the AV confabulates the more specific claim "the passage discusses the Joseon dynasty." Because "Joseon dynasty" is itself strongly associated with Korean history, its recovered direction may overlap substantially with the Korean-history representation in h. Ablating $\widehat{\Delta}_i$ can therefore remove part of that broad theme. This makes the AV less likely to produce C_i even though h never contained enough information to distinguish the claim from other Korean-history-consistent possibilities.

Projection-induced anti-claim signal. Projection ablation can also do more than remove information that was already present. If $\widehat{\Delta}_i$ mixes a broad thematic component with a claim-specific component, projecting the whole direction out of h can introduce a new component pointing *against* the claim-specific direction. The ablated activation may therefore actively suppress C_i, even if the original activation contained no claim-specific signal at all.

A formal derivation of both phenomena is present in Appendix C.

To control for this, we compare each real claim against matched *decoys*: plausible alternative claims with the same broad theme as C_i whose specific content is not supported by the source activation. If ablating a decoy's direction also suppresses the decoy under the AV, that suppression must arise from shared thematic information or the geometry of the intervention, not from specific grounding. The decoys thus provide a baseline against which to measure the real claim's self-ablation effect.

For each real claim C_i, we construct k decoys $C^*_{i,1}, \ldots, C^*_{i,k}$. Each decoy is a sibling claim of the same semantic type and comparable specificity as C_i, but it does not appear in the original AV explanation. Decoys are filtered to match the real claim on four properties that independently affect the self-ablation effect: source-text support, projection magnitude $|h^\top \widehat{\Delta}^*_{i,m}|$, activation-independent baseline likelihood, and splice coherence. Appendix D describes these filters in detail.

We replace C_i with each decoy in the same position in the explanation. Let z_{-i} denote the explanation obtained by deleting C_i, and let $z^*_{i,m}$ denote the explanation obtained by inserting $C^*_{i,m}$ into that same slot.

We then recover a direction for each decoy using the same AR contrast used for the real claim:

$$\begin{aligned} v_{-i} &= \frac{\mathrm{AR}_\theta(z_{-i})}{\|\mathrm{AR}_\theta(z_{-i})\|_2}, \\ v^*_{i,m} &= \frac{\mathrm{AR}_\theta(z^*_{i,m})}{\left\|\mathrm{AR}_\theta(z^*_{i,m})\right\|_2}, \\ \Delta^*_{i,m} &= v^*_{i,m} - v_{-i}, \\ \widehat{\Delta}^*_{i,m} &= \frac{\Delta^*_{i,m}}{\left\|\Delta^*_{i,m}\right\|_2}. \end{aligned} \tag{10}$$

Because the real claim and every decoy are each compared against the same base reconstruction v_{-i}, their recovered directions differ only in which claim fills the slot.

Next, we apply the same causal intervention to each decoy. Let $h^{*\mathrm{abl}}_{i,m}$ denote the activation obtained by projecting the decoy direction out of h:

$$h^{*\mathrm{abl}}_{i,m} = h - \left(h^\top \widehat{\Delta}^*_{i,m}\right) \widehat{\Delta}^*_{i,m}. \tag{11}$$

We first ask how much this intervention suppresses the decoy itself. Using the same scoring procedure as for the original explanation, let the

decoy self-ablation effect be

$$D_{i,m} = S_{z^*_{i,m}}\left(C^*_{i,m} \mid h\right) - S_{z^*_{i,m}}\left(C^*_{i,m} \mid h^{*\mathrm{abl}}_{i,m}\right). \tag{12}$$

$D_{i,m}$ measures how much removing the direction recovered from $C^*_{i,m}$ reduces the AV's score for that same decoy. Because the decoys are ungrounded by construction, their self-ablation effects $D_{i,1}, \ldots, D_{i,k}$ form the matched baseline for the real self-ablation effect R_i, capturing both theme-level piggybacking and the generic anti-claim effect induced by projection.

We also use each decoy intervention to measure its effect on the *real* claim. Specifically, we keep the same ablated activation $h^{*\mathrm{abl}}_{i,m}$, but score C_i in the original explanation z:

$$X_{i,m} = S_z(C_i \mid h) - S_z\left(C_i \mid h^{*\mathrm{abl}}_{i,m}\right). \tag{13}$$

$X_{i,m}$ measures how much ablating a matched decoy direction changes the AV's score for the real claim C_i. Because the intervention removes semantic information shared by the decoy and C_i, while its anti-self component is directed against the decoy rather than C_i, $X_{i,m}$ primarily measures how much C_i depends on the broader shared theme.

After constructing these measurements, we verify that the decoys form a valid negative-control family with the matching filters in Appendix D. If fewer than k valid decoys remain for C_i, we treat the claim as *unresolved*.

2.5 Claim-Level Grounding Classification

Using the decoy family, we classify claims in two stages.

Specificity test. We first ask whether C_i's self-ablation effect exceeds the effects produced by its matched decoys:

$$R_i > Q_{0.95}\left(D_{i,1}, \ldots, D_{i,k}\right) + \kappa_{\mathrm{spec}}. \tag{14}$$

Exceeding the decoy baseline provides evidence that h contains information specific to C_i.

Attribution test. However, passing the specificity test does not show that the C_i-specific information in h is uniquely captured by $\widehat{\Delta}_i$. Closely related claims may have overlapping recovered directions, so ablating another claim's direction may suppress C_i nearly as strongly. Therefore, we test whether C_i's own direction has a substantially larger effect on C_i than any other real claim direction:

$$R_i > \rho_{\mathrm{own}} \max\left\{0,\ \max_{r \neq i} \Delta \log p_{r \to i}\right\}. \tag{15}$$

Here, the parameter $\rho_{\mathrm{own}} > 1$ sets the required margin. In our experiments, $\rho_{\mathrm{own}} = 1.5$.

The specificity test shows that C_i's effect exceeds the matched-decoy baseline, while the attribution test shows that this excess is preferentially associated with $\widehat{\Delta}_i$ rather than other real claim directions. We therefore classify claims that satisfy both tests as *grounded*.

If a claim passes the specificity test but fails the attribution test, we classify it as *entangled*: there is evidence for claim-specific information in the activation, but that information cannot be uniquely attributed to $\widehat{\Delta}_i$.

Distinguishing seeded and confabulated claims. If C_i fails both tests, we find no evidence that h encodes the claim's specific content. However, h may still encode the broader theme that made C_i a plausible continuation for the AV.

To test for this broader thematic support, we use the cross-decoy effects $X_{i,m}$ defined above. Each $X_{i,m}$ measures how much ablating a theme-matched decoy direction suppresses the real claim C_i. We summarize these effects across the decoy family by

$$T_i = \mathrm{median}_m\, X_{i,m}. \tag{16}$$

A positive T_i means that removing directions associated with theme-matched alternatives consistently lowers the AV's score for C_i, providing evidence that h contains broader thematic information that supports the claim. If

$$T_i > \kappa_{\mathrm{theme}}, \tag{17}$$

we classify C_i as *seeded*.

We also measure whether the source activation provides any support for C_i, relative to a generic reference activation. Let h_{ref} denote the mean activation over a large calibration corpus, passed through the same preprocessing and injection pipeline as h.

We define the activation-specific support for C_i as

$$A_i = S_z(C_i \mid h) - S_z(C_i \mid h_{\mathrm{ref}}). \tag{18}$$

A positive A_i means that the actual source activation makes C_i more likely than the generic reference does.

If C_i fails the specificity test and also satisfies

$$T_i \leq \kappa_{\text{theme}} \qquad \text{and} \qquad A_i \leq \kappa_{\text{supp}}, \tag{19}$$

we classify it as *confabulated*. This occurs when we find no evidence that h supports either its specific content or its broader theme, and when the claim is no more likely under the source activation than under the reference baseline.

2.6 Groundedness Results

We evaluate the Qwen2.5-7B layer-20 NLA on 1,500 held-out FineWeb documents. To assess performance in practically relevant settings beyond pretraining-like text, we also evaluate on four representative benchmarks: GPQA Diamond, MMMLU, SWE-bench Verified, and Blackmail.

Half of claims show specific support. On FineWeb, 31% of claims are classified as grounded and 19% as entangled (Table 1). Another 25% of claims are classified as seeded, which is nearly twice the fraction of claims that are classified as confabulated (13%). This suggests that NLAs are often directionally informative, but also frequently introduce unsupported details that are thematically consistent with information present in the activation.

This pattern shifts modestly across evaluations. Specificity rates remain close to FineWeb on GPQA Diamond (49% vs. 50%), but fall on MMMLU (45%), SWE-bench Verified (42%), and Blackmail (41%). SWE-bench and Blackmail also produce more unresolved claims (18% and 17%, vs. 12% on FineWeb), suggesting that the grounding test is less decisive in some out-of-distribution settings.

Decoys improve on claim deletion. Does the test identify supported details more reliably than reconstruction alone? We construct 300 pairs of contexts that differ in one attribute, such as a person's occupation, while preserving the topic. We score the same claim under both activations and test whether each method distinguishes source-consistent from source-inconsistent claims. Comparisons use the same 1,056 scorable cases out of 1,200 attempted claim–activation cases.

Our main baseline is the increase in reconstruction error after deleting a claim, as evaluated in the original NLA paper (Fraser-Taliente et al., 2026). We compare this with the decoy-corrected self-ablation score:

$$\begin{aligned} B_i^{\text{del}} &= E_{\text{dir}}(z_{-i}; h) - E_{\text{dir}}(z; h), \\ G_i &= R_i - Q_{0.95}(D_{i,1}, \ldots, D_{i,20}), \end{aligned} \tag{20}$$

where E_{dir} is squared error between unit-normalized target and reconstructed activations. Higher scores indicate stronger support.

The corrected score reaches .82 AUROC, compared with .72 for claim deletion (paired difference .10, 95% CI [.06, .14]; Table 2). Without decoy correction, self-ablation performs similarly to claim deletion (.73 AUROC) and has a higher false-positive rate (24% versus 12%). Correction reduces this rate to 11%. The improvement therefore comes from comparing a claim with plausible alternatives, not merely from measuring the AV's response to an intervention.

This benchmark tests sensitivity to an explicit change in the source. It does not establish that the AV faithfully reports latent beliefs or other information that cannot be checked against the input.

SAEs distinguish themes from details. We next audit 200 natural claims from each category against an SAE at the same model and activation site. Feature annotations are assessed without access to our classifications. We record support for the exact claim and its broader theme separately. Support rates are computed only where the dictionary contains an independently validated feature for the claim; missing features are not counted as evidence against it.

SAEs support the exact content of 82% of grounded claims and 75% of entangled claims (Table 3). Entanglement therefore need not indicate an unsupported claim. For seeded claims, thematic support remains high (83%), but exact support falls to 25%. This is the distinction the decoy controls are intended to capture: evidence for a topic need not support the particular detail chosen by the AV.

SAEs are also a strong verifier on the controlled benchmark. On the 660 cases with feature coverage, SAE support reaches .85 AUROC versus .83 for our score (paired difference $-.02$, 95% CI $[-.06, .02]$). We find no advantage over SAEs on this subset. The proposed test covers additional claims, although that broader coverage does not itself establish their reliability. Finally, 20% of covered claims classified as confabulated have exact SAE support. These disagreements show why a negative result from the test should not be treated as proof that the activation lacks the claimed information.

Category	FineWeb	GPQA Diamond	MMMLU	SWE-bench Verified	Blackmail
Grounded	31.0 [28.9, 33.2]	29.0 [22.4, 36.5]	27.0 [20.3, 34.6]	22.0 [15.8, 29.2]	23.0 [16.6, 30.4]
Entangled	19.0 [17.3, 20.8]	20.0 [14.7, 26.3]	18.0 [12.8, 24.2]	20.0 [14.2, 26.8]	18.0 [12.6, 24.5]
Seeded	25.0 [23.0, 27.1]	25.0 [19.0, 31.8]	24.0 [18.0, 31.0]	26.0 [19.5, 33.4]	29.0 [22.1, 36.7]
Confabulated	13.0 [11.5, 14.6]	14.0 [9.5, 19.5]	15.0 [10.1, 20.9]	14.0 [9.1, 20.0]	13.0 [8.4, 18.7]
Unresolved	12.0 [10.6, 13.6]	12.0 [7.8, 17.3]	16.0 [10.9, 22.0]	18.0 [12.5, 24.6]	17.0 [11.6, 23.5]

Table 1: Claim classifications (%) with 95% intervals.

Score	AUROC	False-positive rate (%)	AUROC, SAE-covered
Whole-explanation reconstruction	.60 [.56, .64]	10	.61 [.56, .66]
Claim-deletion reconstruction	.72 [.68, .76]	12	.74 [.69, .79]
Activation-specific support, A_i	.74 [.70, .78]	13	.76 [.71, .81]
Self-ablation, R_i	.73 [.69, .77]	24	.75 [.70, .80]
Decoy-corrected self-ablation, G_i	.82 [.78, .86]	11	.83 [.78, .88]
SAE support	—	—	.85 [.81, .89]

Table 2: Discrimination between source-consistent and source-inconsistent claims.

Category	Covered	Exact (%)	Theme (%)
Grounded	150/200	82.0	92.0
Entangled	140/200	75.0	90.7
Seeded	120/200	25.0	83.3
Confabulated	90/200	20.0	35.6
Unresolved	100/200	41.0	62.0

Table 3: Results. Active SAE support among claims with validated dictionary coverage. Exact and thematic support can overlap.

3 Robustness

3.1 Robustness to the Warm Start

A central assumption behind NLA training is that the unsupervised reconstruction objective eventually makes the natural-language bottleneck reflect information in the activation, rather than arbitrary features of its supervised initialization. If this is true, then sufficiently trained NLAs should converge toward similar explanations even when they begin from substantially different warm starts.

Recent results cast doubt on this assumption. Zhang and Turner train Qwen2.5-7B NLAs using the standard NLA pipeline, but replace the usual teacher-generated warm-start descriptions with deliberately implausible ones. Despite this change, the resulting NLA recovers nearly the same reconstruction accuracy as a normally initialized control: after RL, the two models reach approximately 0.68 and 0.70 FVE, respectively. However, the implausible-initialized NLA continues to produce overwhelmingly implausible claims. This demonstrates that high reconstruction accuracy alone does not guarantee plausible natural-language explanations.

Plausibility, however, does not directly test whether reconstruction training selects a stable interpretation of the activation. Two NLAs could disagree substantially while both producing plausible explanations, or they could produce the same implausible claims because those claims encode information useful to the AR. We therefore ask a stronger question: *when NLAs are trained from radically different warm starts but reach similar reconstruction accuracy, do they converge on the same semantic claims?*

Training conditions. We follow the initialization-perturbation procedure of Zhang and Turner. For each training seed, we train two NLAs on the same target model, activation layer, training examples, and reconstruction objective. The *standard-initialized* NLA receives the usual warm start, in which a teacher model describes features that are plausibly relevant for predicting the continuation of each text prefix. The *implausible-initialized* NLA instead receives teacher descriptions generated under an instruction to make the corresponding claims false while remaining related to the source. All subsequent NLA training is identical between the two conditions. In particular, both AVs are optimized against the same reconstruction objective and both ARs are trained on the same underlying activation distribution.

We evaluate both the final checkpoints and checkpoints matched as closely as possible in FVE.

Cross-initialization claim agreement. For a held-out activation h, let z^{std} and z^{imp} denote explana-

tions sampled from the standard- and implausible-initialized AVs. We decompose each explanation into atomic claims using the same claim-segmentation procedure described in Appendix A. We then use a separate LLM judge to determine whether claims from the two explanations express the same semantic content. We match claims by semantic equivalence at the same level of specificity: two claims are counted as the same if they express the same underlying proposition, allowing for paraphrase or minor differences in wording, but not if one is merely topically related, more general, or more specific than the other.

Because the two explanations may contain different numbers of claims, we score overlap in both directions. Let $C^{(a)}$ and $C^{(b)}$ denote the claim sets extracted from the two explanations. We define

$$P_{a\to b} = \frac{\left|\left\{ \begin{array}{c} c \in C^{(a)} : \exists\, c' \in C^{(b)}, \\ \text{such that } c \equiv c' \end{array} \right\}\right|}{|C^{(a)}|}, \quad (21)$$

where $c \equiv c'$ denotes semantic equivalence under the matching criterion above. We define $P_{b\to a}$ analogously.

We then summarize the two directional overlap rates with their harmonic mean:

$$A_{\text{cross}} = \frac{2P_{a\to b}P_{b\to a}}{P_{a\to b} + P_{b\to a}}. \quad (22)$$

This penalizes cases in which one explanation is largely contained within the other but the reverse is not true. We refer to A_{cross} as *cross-initialization claim agreement*.

We compare this against repeated samples from the same AV on the same activation, using the default sampling temperature of 0.7. This gives a baseline for how much explanations vary from sampling alone. We then ask whether agreement across different warm starts is close to this level. We also report agreement separately for broad thematic claims and for more specific entity- and detail-level claims.

Evaluation distributions. We first measure claim agreement on held-out pretraining-like text from FineWeb, which is close to the distribution used to train the NLA. This tests whether the reconstruction objective converges on a common semantic code in the setting most favorable to the method.

We then repeat the analysis on out-of-distribution activations from the 16 evaluations used in the original NLA study's evaluation-awareness analysis: seven alignment evaluations (Blackmail, Destructive Coding Actions, Overly Agentic Actions, Factual Honesty, Appropriate Harmlessness, Sycophancy, and Multi-turn Deception) and nine capability benchmarks (SWE-bench Verified, GPQA Diamond, Humanity's Last Exam, τ^2-bench Retail, MMMLU, BrowseComp, Terminal-Bench 2, FinanceAgent, and CyberGym).

3.2 Robustness of the AV–AR Pipeline

Our second experiment asks whether the AR reconstructs activations from the *semantic content* of an AV explanation, or whether reconstruction depends on more superficial properties of the text produced by the AV.

This question is motivated by a recurring observation about the contents of NLA explanations. Alongside high-level descriptions of the activation, the AV often produces quoted or near-quoted text resembling tokens from the surrounding input. In some cases these quotes reproduce the source closely; in others they resemble the source while changing words, entities, or details. This may simply be a convenient way for the AV to describe what the target model is representing. Alternatively, it may reflect *input inversion*: the AV may be reconstructing parts of the source text, which the AR—a fine-tuned copy of the target model—can then effectively run again to reproduce h. In that case, reconstruction may succeed because the AV recovers the input that generated the activation, rather than because it faithfully describes the activation's contents.

At a high level, if the AR primarily uses the meaning of an explanation, then meaning-preserving edits should leave reconstruction largely intact. If instead the AR relies on text that resembles the target model's original input, then reconstruction may be unusually sensitive to quoted or input-like spans, even when their semantic content is preserved.

We test this by applying controlled transformations to held-out AV explanations. For an original explanation z, let

$$\hat{h} = \text{AR}_\theta(z)$$

denote its reconstruction. For each transformation f, we construct an edited explanation

$$z^{(f)} = f(z)$$

and measure the resulting change in reconstruction

error,

$$\Delta E_f = \left\| h - \mathrm{AR}_\theta(z^{(f)}) \right\|_2^2 - \|h - \mathrm{AR}_\theta(z)\|_2^2 . \tag{23}$$

Positive ΔE_f indicates that the transformation harms reconstruction; negative values indicate that the edited explanation reconstructs the source activation better than the original AV output. We also report the corresponding change in FVE over the evaluation set.

We evaluate five transformations.

Removing quoted content. We first remove all quoted spans from the AV explanation while leaving the remaining text unchanged. This measures how much reconstruction information is carried by quoted material as a whole. Because this removes the quoted content entirely, it does not tell us whether the AR depended on the wording of the quote or on the information it contained. It is therefore a coarse measure of how much the quoted spans matter for reconstruction.

Removing input-redundant quotes. Next, we only remove quoted spans that reproduce information already present in the source text. This includes both verbatim quotations and close paraphrases that restate the source without adding a distinct claim about the activation. Quoted spans that differ substantively from the source are retained. This tests how much reconstruction depends specifically on the AV reproducing an approximate copy of the original input.

We identify input-redundant quotes using two criteria:

1. **Deterministic string match.** The quoted span contains a contiguous verbatim sequence of at least two content words from the source.

2. **Semantic support.** Claude Haiku judges the quoted content to be explicitly stated by, or directly entailed by, the source text.

Paraphrasing quoted content. To separate the information in a quote from its exact wording, we rewrite each quoted span as ordinary descriptive prose while preserving its meaning. For example, "the cabinet meeting on the following day" might become "the text concerns a cabinet meeting occurring the next day." This keeps the underlying information while removing both the quotation format and the close wording match to the source. If reconstruction remains stable, the AR is mainly using the information conveyed by the quote. If it drops sharply, the AR may instead rely on the quote's source-like wording or structure.

Replacing AV quotes with source text. For quoted text that are classified as reconstructions of the preceding context, we replace the AV's version with the exact source text while leaving the rest of the explanation unchanged. For each selected quote, we take the same number of whitespace-delimited words from the end of the source prefix, up till the activation's read position, and recompute the reconstruction. For example, if the AV writes ''the cabinet meeting took place the next morning", and the source prefix ends with "the cabinet meeting convened early the following day," we replace the AV quote with those final source words and recompute the reconstruction.

This tests whether the AV's deviations from the source are simply imperfect copies or whether they carry information about h. If the exact source text reconstructs better, the AV's altered wording is likely reconstruction noise; if it reconstructs worse, the deviation may better reflect how the source is represented in the activation.

Reordering claims. We randomly permute the substantive claims in each explanation while leaving the claims themselves unchanged. This tests whether the AR primarily uses the set of semantic claims in z, or whether reconstruction depends strongly on their original ordering and local textual context.

We randomly permute the contents of each explanation while leaving its substantive claims unchanged, testing whether the AR primarily relies on the set of semantic claims in z or also on their ordering and local textual context. We consider three levels of permutations:

1. **Block shuffling.** Randomly reorders paragraph-sized blocks of the explanation.

2. **Sentence shuffling.** Randomly reorders sentences while preserving quoted spans.

3. **Quoted-claim shuffling.** Shuffles only the quoted predictions about what text comes next within the final block, while leaving the quoted read token in its original position. For example, if the block predicts "tools," "Generator

templates," and "scripts," we randomly reorder those three quotations but leave the quoted read token "Generator" fixed.

The final intervention most directly isolates sensitivity to the ordering of individual claims.

Results. We evaluate each manipulation on $n = 2{,}000$ held-out documents per model. We report 95% document-bootstrap percentile confidence intervals using 20,000 resamples. For each manipulation, we test $\Delta\mathrm{FVE} = 0$ with a two-sided paired t-test ($df = 1{,}999$) and apply a Benjamini–Hochberg correction across the 48 manipulation tests. All effects reported below remain significant after correction ($q < 10^{-3}$).

The largest effects come from manipulating quoted content. Removing all quoted spans reduces FVE by 0.73–0.99, nearly eliminating reconstruction for both Gemma models and pushing Qwen below the mean-prediction baseline. Paraphrasing the quoted spans while preserving their meaning also produces a large drop, reducing FVE by 0.28–0.39. Thus, quoted spans matter not only for the information they contain: the AR is also sensitive to how that information is expressed.

By contrast, removing only input-redundant quotations has a much smaller effect, reducing FVE by 0.03–0.08. Reordering claims is similarly modest for most model–manipulation pairs, although quoted-claim shuffling has a larger effect for Qwen2.5-7B. Most notably, replacing the AV's approximate reconstructions of prior context with exact source text slightly *improves* reconstruction for all three models ($\Delta\mathrm{FVE} = +0.004, +0.019, +0.012$, respectively). This is consistent with input inversion: source-like text appears to give the AR a particularly effective route for reconstructing the original activation.

Figure 3 shows the change in FVE for every manipulation, while Figure 4 reports the corresponding absolute reconstruction quality.

Manipulation	Qwen2.5-7B	Gemma-3-12B	Gemma-3-27B
Baseline FVE	0.748 [0.744, 0.752]	0.750 [0.742, 0.757]	0.768 [0.760, 0.775]
Remove all quoted content	-0.991 [-1.002, -0.981]	-0.734 [-0.748, -0.719]	-0.823 [-0.848, -0.797]
Remove input-redundant quotes (string)	-0.041 [-0.045, -0.036]	-0.034 [-0.038, -0.030]	-0.072 [-0.080, -0.065]
Remove input-redundant quotes (semantic)	-0.032 [-0.036, -0.027]	-0.034 [-0.039, -0.030]	-0.076 [-0.084, -0.069]
Paraphrase quoted content	-0.392 [-0.403, -0.380]	-0.293 [-0.304, -0.283]	-0.277 [-0.288, -0.266]
Block shuffle	-0.051 [-0.055, -0.048]	-0.018 [-0.020, -0.017]	-0.023 [-0.024, -0.021]
Sentence shuffle	-0.040 [-0.042, -0.037]	-0.018 [-0.019, -0.017]	-0.023 [-0.025, -0.021]
Quoted-claim shuffle	-0.112 [-0.120, -0.105]	-0.031 [-0.033, -0.029]	-0.021 [-0.023, -0.020]
Replace AV quotes with source text	+0.004 [+0.002, +0.006]	+0.019 [+0.015, +0.022]	+0.012 [+0.010, +0.013]

Table 4: Reconstruction robustness across three open-model NLAs.

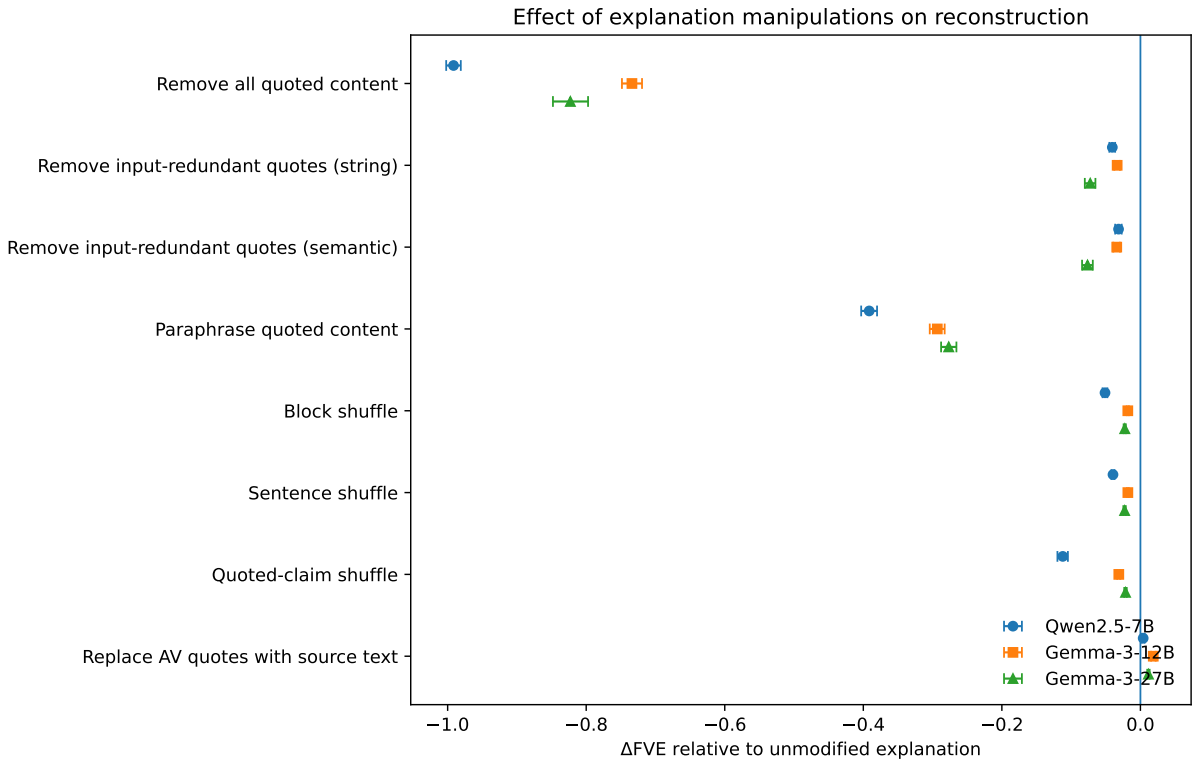


Figure 3: Effect of each explanation manipulation on reconstruction. Points show ΔFVE relative to the unmodified explanation; error bars are 95% document-bootstrap confidence intervals.

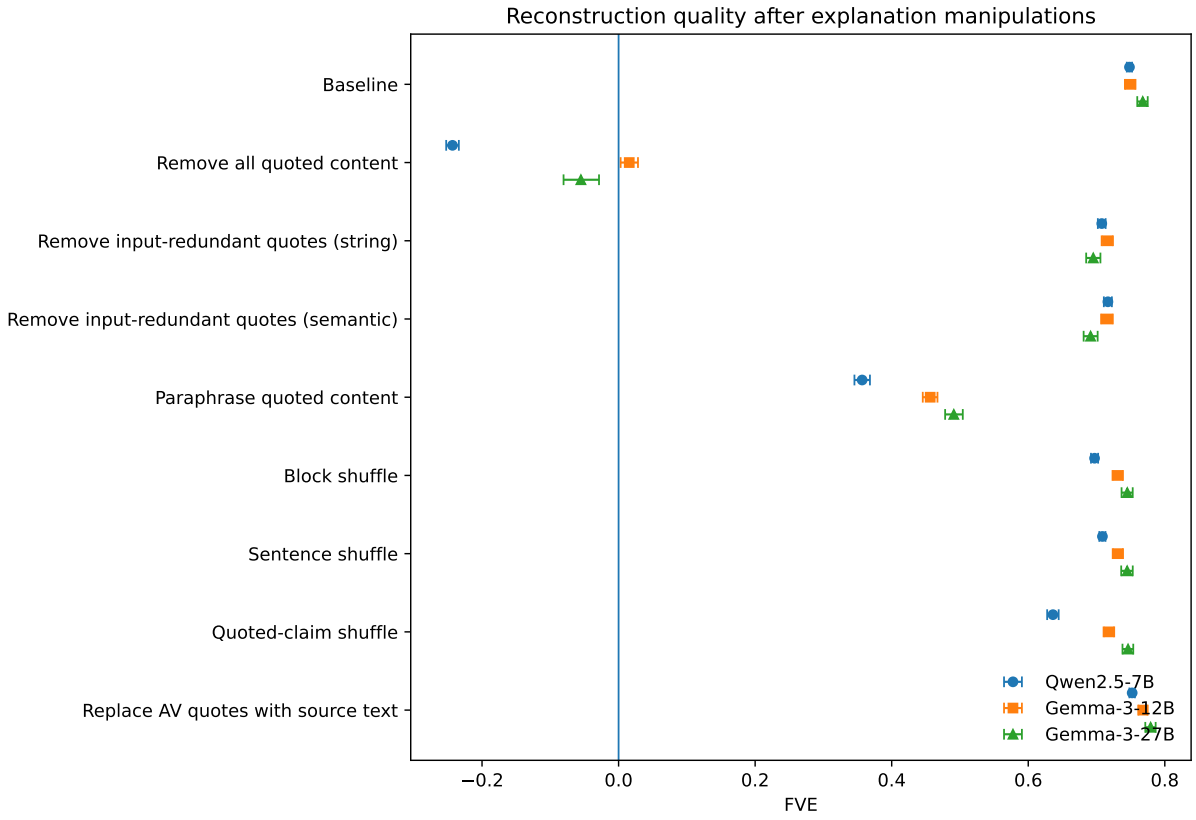


Figure 4: Absolute reconstruction quality for the baseline and every explanation manipulation. Points show FVE; error bars are 95% document-bootstrap confidence intervals.

Appendix

A Claim Segmentation

Before constructing claim directions, we partition each AV explanation into discrete claims using Claude Sonnet 4.6 at temperature 0. The goal is to obtain atomic units that can be deleted and scored independently while preserving an exact partition of the original explanation.

1. **Atomic unit.** A claim is exactly one assertion about the source. Claims therefore need not correspond to complete sentences or syntactic phrases.

2. **Claim boundaries.** We require each claim to be expressible as a single sentence of the form "The source <is / is about / reads as / contains / expects> ___," making exactly one assertion. We use this as the boundary test:

   - If a span contains two independent assertions, we split it.
   - If splitting a span would change or destroy the meaning of the assertion, we keep it intact.

3. **Connectives.** Text that carries no assertion of its own, such as punctuation, whitespace, or bare glue words such as "and," "or," or "from," is labeled a *connective*. Connectives are retained so that the segmentation preserves every character of the original explanation, but are excluded from all subsequent claim-level analyses.

   For example, the sentence "News report about a Georgia shooting, establishing a factual narrative about a firearms conviction" is segmented as "News report about a Georgia shooting" / "," / "establishing a factual narrative" / "about a firearms conviction." The three substantive spans are claims; the comma is a connective.

4. **Exact tiling.** The resulting segments must form a flat, non-overlapping partition of the explanation. Concatenating their text fields in order must reproduce the original explanation character-for-character, including whitespace, punctuation, quotation marks, and newlines. Thus every character belongs to exactly one claim or connective, with no overlap or nesting.

Edge cases. We apply two additional rules:

1. **Quoted strings.** A quoted string is treated as a separate claim. A quote together with a short phrase making a single assertion about that quote may instead be kept as one claim.

2. **Hedges.** Hedge words such as "likely," "probably," and "expecting" are part of the assertion they qualify and therefore remain attached to the corresponding claim rather than being labeled as connectives.

B NLA Vector Injection and Activation Normalization

To query the NLA, we first extract a residual-stream activation $h_{\ell,t} \in \mathbb{R}^{d_{\text{model}}}$ from layer ℓ and token position t of the target model. The activation is L_2-normalized,

$$\bar{h}_{\ell,t} = \frac{h_{\ell,t}}{\|h_{\ell,t}\|_2},$$

and then rescaled by a fixed scalar α. The resulting vector

$$h_{\text{inj}} = \alpha \bar{h}_{\ell,t}$$

is inserted into the activation verbalizer (AV) by **replacing the embedding of a designated special token** in a fixed prompting template. Thus, the AV receives an ordinary sequence of token embeddings except at one position, where the token embedding is replaced by the target-model activation. It then autoregressively generates a natural-language description of that activation (Fraser-Taliente et al., 2026).

For the grounding experiments, the symbol h denotes $\bar{h}_{\ell,t}$. We apply the projection ablations in Eqs. 6 and 11 to this normalized vector, then multiply both the original and projected vectors by the same α before injection. We do not apply a second normalization after the projection; doing so would change the intervention defined by those equations.

Normalizing the original activation to unit norm makes the NLA insensitive to variation in its overall magnitude and stabilizes reconstruction training; the injected vector's initial norm is controlled by α. For the released open-model NLAs, α is fixed

throughout training and is chosen to be relatively large. A practical heuristic used by the authors is

$$\alpha \approx \mathrm{Quantile}_{0.75}\left(\left\{\|h_{\ell,t}\|_2 : (x,t) \sim \mathcal{D}\right\}\right),$$

i.e. approximately the 75th-percentile norm of raw activations at the target layer over a large corpus. The authors hypothesize that this large injected norm helps the activation survive the AV's early layers relatively unchanged until it reaches layers capable of interpreting a representation from the target depth ℓ (Fraser-Taliente et al., 2026).

C Why Self-Ablation Alone Does Not Establish Groundedness

A positive self-ablation effect shows that removing the direction recovered from a claim makes the AV less likely to reproduce that claim. However, this does not by itself show that the claim's specific content was encoded in the original activation. The same effect can arise even when the activation contains only a broader semantic theme related to the claim.

To see this, consider a stylized model of a single claim C_i. Let u be a unit direction representing the claim's broad semantic theme, and let s be a unit direction representing the claim's specific content. Assume that these directions are orthogonal:

$$u^\top s = 0, \qquad \|u\|_2 = \|s\|_2 = 1. \tag{24}$$

Suppose the unit direction recovered from the AR mixes these two components:

$$\widehat{\Delta}_i = au + bs, \qquad a^2 + b^2 = 1, \qquad a, b > 0. \tag{25}$$

Here, a is the component of the recovered direction along the broader theme, and b is its component along the claim-specific content.

Now consider the null case in which the specific content of C_i is not encoded in the source activation. Let

$$h = cu + r, \qquad c > 0, \tag{26}$$

where r is orthogonal to both u and s. Thus, h contains the broad theme with strength c, but has exactly zero component along the claim-specific direction s.

Because

$$h^\top \widehat{\Delta}_i = ca, \tag{27}$$

the projection ablation in Eq. 6 gives

$$\begin{aligned} h_i^{\mathrm{abl}} &= h - (ca)\widehat{\Delta}_i && (28)\\ &= cu + r - ca(au + bs) && (29)\\ &= c(1-a^2)u - cabs + r && (30)\\ &= cb^2 u - cabs + r. && (31) \end{aligned}$$

Although the original activation contains no claim-specific component, the ablation changes it in two ways: it weakens the genuine theme signal and creates a new component pointing opposite to the claim-specific direction.

Theme piggybacking. A confabulated specific claim can still depend on genuine broader information in the activation. For example, the activation might encode "Korean history," while the AV confabulates the more specific but unsupported claim "Joseon Dynasty." Because that claim is semantically related to Korean history, its AR-derived direction can have a large component along u. The ablation then reduces the theme coefficient from c to cb^2. If the Korean-history signal made "Joseon Dynasty" a plausible completion, weakening that signal can lower the AV's score for the claim even though the activation never contained enough information to distinguish Joseon from other Korean-history-consistent possibilities.

Projection-induced anti-claim signal. Equation 31 also contains the term $-cabs$. The original activation had zero component along s, but the ablated activation has a negative component along it. The projection can therefore do more than remove information that was originally present: it can move the activation in a direction that actively suppresses the claim.

We can express both effects directly in terms of the claim score. Define

$$L_i(x) = S_z(C_i \mid x). \tag{32}$$

For a sufficiently small intervention, a first-order expansion around h gives

$$L_i(h_i^{\mathrm{abl}}) \approx L_i(h) + \nabla L_i(h)^\top (h_i^{\mathrm{abl}} - h). \tag{33}$$

Therefore,

$$\begin{aligned} R_i &= L_i(h) - L_i(h_i^{\mathrm{abl}}) && (34)\\ &\approx \nabla L_i(h)^\top (h - h_i^{\mathrm{abl}}) && (35)\\ &= ca\,\nabla L_i(h)^\top (au + bs). && (36) \end{aligned}$$

Let

$$g_{\text{theme}} = \nabla L_i(h)^\top u, \qquad g_{\text{specific}} = \nabla L_i(h)^\top s. \tag{37}$$

Then

$$R_i \approx ca^2 g_{\text{theme}} + cab g_{\text{specific}}. \tag{38}$$

The first term captures the decrease in claim score caused by removing genuine theme information. The second captures the decrease caused by introducing a component opposite to the claim-specific direction. Both terms can be positive even though the original activation has zero component along s.

Thus, a positive R_i does not by itself establish that the specific content of C_i was encoded in h. This motivates the matched decoy control in the main experiment. For each real claim, we construct alternative claims with the same broader theme but unsupported specific content, and apply the same AR reconstruction, projection ablation, and AV rescoring procedure. Their self-ablation effects $D_{i,m}$ estimate how much suppression can arise from theme overlap and from the geometry of the intervention itself. Evidence for specific grounding therefore requires R_i to exceed this matched-decoy baseline.

The two-dimensional decomposition above is deliberately simplified. In practice, activations are high-dimensional, themes and claims need not correspond to orthogonal directions, and the linear approximation may break down for large interventions. The example is meant only to show that the self-ablation statistic can be positive even when the source activation contains no claim-specific information.

D Decoy Construction and Filtering

The purpose of the decoy family is to construct a matched reference condition for each real claim. A useful decoy should preserve properties that can produce a positive self-ablation effect even in the absence of specific grounding, while replacing the particular content whose grounding we wish to test. We therefore generate a larger pool of candidate decoys and retain only candidates that satisfy the criteria below.

Claim type. We first classify each real claim according to the kind of information it asserts. We use Claude Sonnet 4.6 at temperature 0 to assign a coarse class and subtype. Theme-level claims are assigned one of

$$\{\text{genre, topic, entity, structure, era, other}\},$$

while detail-level claims are assigned one of

$$\{\text{quote, date, number, value, other}\}.$$

Candidate decoys are generated as sibling claims with the same class and subtype as the real claim. The goal is to change the substantive content of the claim while preserving its approximate level of specificity and its role in the explanation. Candidates that reproduce a claim already present in the original AV explanation are discarded.

For every candidate $C^*_{i,m}$, we insert the candidate into the position originally occupied by C_i, forming $z^*_{i,m}$, and compute its direction using Eq. 10. We then apply four additional filters.

1. Absence from the source text. The specific content of a decoy should not have independent support from the example being interpreted. Otherwise, the decoy may itself be grounded and would no longer provide an appropriate control. We therefore reject candidates that are stated, paraphrased, or directly entailed by any span of the source text.

This criterion does not prove that a decoy is absent from the activation: the target model may infer information that is not stated explicitly in its input. Rather, it removes candidates for which we have direct observable evidence of support and therefore prevents obviously grounded claims from entering the control set.

2. Matched intervention magnitude. The projection ablation associated with a unit direction $\widehat{\Delta}$ removes

$$\left(h^\top \widehat{\Delta}\right) \widehat{\Delta} \tag{39}$$

from the source activation. Its magnitude is therefore

$$\left|h^\top \widehat{\Delta}\right|. \tag{40}$$

This quantity matters independently of the semantic content of the direction: a decoy whose direction has little overlap with h receives a much weaker intervention than the real claim and is therefore not a comparable control.

We consequently require the magnitude of the decoy projection to be close to that of the real claim:

$$\left|\left|h^\top \widehat{\Delta}^*_{i,m}\right| - \left|h^\top \widehat{\Delta}_i\right|\right| \le \varepsilon_{\text{ov}} \left|h^\top \widehat{\Delta}_i\right|. \tag{41}$$

We choose the initial tolerance ε_{ov} on a calibration split. If fewer than k candidates survive, we widen the tolerance according to a fixed schedule until enough decoys remain.

3. Matched activation-independent propensity. The magnitude of a log-probability change also depends on how strongly the AV already favors a particular sequence of tokens. A highly probable claim and an otherwise comparable but extremely unlikely claim need not respond similarly to the same-sized activation perturbation. We therefore match decoys to the real claim on their baseline propensity under the AV.

Let h_{ref} be the fixed reference activation defined in Section 2.5. Because the same reference is used for every example, scoring under it removes dependence on the particular source activation while preserving the AV's generic lexical and discourse preferences. Since the claim score in Eq. 7 is already normalized by token length, we retain a candidate only if

$$\left| S_{z^*_{i,m}}(C^*_{i,m} \mid h_{\text{ref}}) - S_z(C_i \mid h_{\text{ref}}) \right| \leq \varepsilon_{\text{base}}. \tag{42}$$

4. Naturalness of the splice. Finally, replacing C_i with a decoy should not itself move the AR far outside the distribution of verbalizer-style explanations on which it was trained. If $z^*_{i,m}$ is ungrammatical or locally incoherent, its recovered direction may reflect this distribution shift rather than the semantic content of the replacement.

We check for this in two ways. First, an LLM judge compares $z^*_{i,m}$ with z and rejects replacements that substantially degrade fluency, discourse coherence, or structural well-formedness. Second, we require the AR's reconstruction error on the edited explanation to remain close to that of the original explanation. We define the per-example reconstruction error

$$E(z; h) = \|h - \text{AR}_\theta(z)\|_2^2 . \tag{43}$$

We retain a candidate only if

$$\left| E(z^*_{i,m}; h) - E(z; h) \right| \leq \varepsilon_{\text{rec}} E(z; h). \tag{44}$$

This rejects splices for which the AR reconstruction changes dramatically for reasons unrelated to the intended semantic substitution.

Decoy-family validity. After applying these filters, we verify that each remaining decoy is a valid negative control: its score under the AV should not increase when conditioned on the source activation relative to its activation-independent baseline. For each decoy, we therefore compute its activation-specific support

$$A^*_{i,m} = S_{z^*_{i,m}}(C^*_{i,m} \mid h) - S_{z^*_{i,m}}(C^*_{i,m} \mid h_{\text{ref}}). \tag{45}$$

A large positive value means that the AV assigns substantially more probability to the decoy when conditioned on h than when conditioned on the reference activation, suggesting that the decoy's specific content may itself be represented in h. We reject candidates whose $A^*_{i,m}$ exceeds a threshold fixed on the calibration split.

E Replacing AV Quotes with Source Text

This ablation tests whether deviations between an AV's quoted reconstruction and the original source carry information about the activation beyond the source content itself. We construct edited explanations in which quotations interpreted as reconstructions of the preceding input are replaced with text drawn directly from the source, while leaving the remainder of the AV explanation unchanged. We then pass the edited explanation to the AR and measure reconstruction against the same original activation. Thus, the intervention changes only the wording supplied inside selected quotations; the reconstruction target is held fixed.

Anchoring the source context. For each example, we recover the exact source position associated with the activation from the token index stored when the activation was collected. We use the tokenizer's offset mapping to convert this token index into character offsets in the original source. This avoids locating the read token by string search, which would be ambiguous when the same token occurs multiple times in a document.

We consider two source prefixes. In the first variant, the prefix extends through the token at which the activation was read. In the second, it ends immediately before that token. Importantly, both variants are evaluated against the *same* original activation; excluding the read token from the replacement text does not move the activation readout to the preced-

ing token. This allows us to distinguish information recoverable from the preceding context from information supplied by explicitly reproducing the read token itself.

Selecting quotations for replacement. We identify candidate quotations using a deterministic parser over the AV explanation. The explanation is first divided into blocks at blank lines, and quoted spans are extracted independently within each block. The parser then assigns quotations to functional categories using their structural position and nearby lexical cues.

In particular, quotations describing the document's genre are excluded, as is the quotation that directly echoes the final/read token. Quotations appearing after phrases that introduce predicted continuations—e.g., 'expecting," 'followed by," 'likely," or 'continuation"—are treated as predictions about future text and are also excluded. Remaining substantive quotations are treated as candidate reconstructions of the left context. Quotes that simply duplicate the read token or contain no qualifying content words are left unchanged.

As an additional safeguard, we compare the content words in each candidate quote with local windows before and after the read token. A quote is reassigned from left-context reconstruction to predicted continuation when its lexical overlap is substantially greater with the future window than with the preceding context. This procedure is deliberately rule-based: no language model is used to determine whether a quotation is semantically faithful to the source.

Constructing the source replacement. For each selected quotation, let n be its number of whitespace-delimited words. We replace its contents with the final n words of the relevant source prefix:

$$q' = \mathrm{suffix}_n(x_{\leq t}),$$

for the variant whose source prefix includes the read token, and analogously with $x_{<t}$ for the variant excluding it. The quotation marks themselves are preserved.

Consequently, different quotations within the same explanation may be replaced by overlapping source spans: every replacement terminates at the same source boundary, while its start position is determined by the length of the original quotation. This approximately holds quotation length fixed while replacing the AV's reconstructed wording with source-local wording. The procedure does *not* search the document for the passage most semantically similar to each quotation, select clause or sentence boundaries, or rewrite the surrounding explanation to improve grammatical fit.

For example, suppose a selected AV quote contains six words and the source prefix ends in "…using Sketch slices or Adobe Generator." The replacement is simply the final six whitespace-delimited words of that prefix, regardless of which particular words or concepts appeared in the original AV quotation.

Editing and scoring. All replacements are applied directly to character spans in the original AV explanation. Each experimental variant is generated independently from that original explanation, rather than by successively editing another variant. Text outside the selected quoted spans is preserved exactly.

The resulting edited explanation is then supplied to the trained AR, which reconstructs an activation vector from the explanation. Reconstruction is evaluated against the original activation collected at the fixed read position. We therefore compare

$$\mathrm{AR}(e_{\mathrm{orig}}) \quad \text{and} \quad \mathrm{AR}(e_{\mathrm{source}})$$

against the same target h, where e_{source} differs from e_{orig} only in the selected quoted spans.

Interpretation. The comparison asks whether the AV-specific wording of reconstructed context contributes information beyond directly supplying nearby source text. If replacing an AV quotation with the corresponding source-local context improves reconstruction, then the AV's deviations from the input are unnecessary or detrimental for reconstructing h. If reconstruction instead worsens, then those deviations carry information useful to the AR that is absent from a direct copy of the source context. Such information could reflect how the input is encoded or transformed in the activation, rather than merely which input tokens preceded it.

The ablation should therefore be interpreted as a comparison between *AV-generated reconstructions of prior context* and *length-matched, source-local text*, rather than as a test in which each AV paraphrase is aligned to and replaced by its unique semantic counterpart in the source.

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